Calculus as the Geometry of Relationship and Controlled Change

A Structural Textbook on Differential and Integral Theory

Flyxion

February 27, 2026

Abstract

This text develops calculus as a unified theory of relational deformation under controlled perturbation. Rather than presenting differentiation and integration as computational techniques, we interpret them as operators governing how structured systems respond to infinitesimal variation and how local influences accumulate into global behavior. The *derivative* is introduced as a sensitivity operator mapping perturbations to structured responses; the *integral* is presented as the reconstruction of coherent form from infinitesimal contributions.

We integrate classical limit theory, multivariable calculus, manifold geometry, dynamical systems, stability analysis, control theory, and functional simulation architecture into a single structural narrative. Tangent and normal decomposition, curvature, and projection are treated not as advanced topics but as intrinsic features of differential reasoning from the outset.

The text further interprets calculus pedagogically as the disciplined management of abstraction under computational constraints, distinguishing structural invariants from arithmetic turbulence. Applications to physiological cell modelling, engineered appliances, and inferential systems demonstrate that the same geometric architecture governs all three domains. Throughout, calculus is framed as the geometry of relationship: a study of how quantities co-vary, how systems stabilise, and how change propagates through interconnected variables.

Contents

1 Change Before Computation **9**

1.1 The Problem of Variation 9

1.2 Rise, Run, and the Balance of Error 10

1.3 Limits as Stability Under Refinement 10

1.4 The Derivative as a Sensitivity Operator 12

1.5 Abstraction as Reduction of Irrelevant Degrees of Freedom 13

2 Higher-Order Change and Curvature **14**

2.1 The Second Derivative and the Geometry of Bending 14

2.2 Critical Points and Stability 15

2.3 From One Variable to Many 15

2.4 Directional Derivatives and Linear Structure 16

3 Tangent Spaces and Local Models **18**

3.1 Curves and Embedded Surfaces 18

3.2 Functions Between Manifolds and the Jacobian 19

3.3 The Chain Rule as Composition of Deformation 19

3.4 The Hessian and Second-Order Structure 20

4 Integration and Reconstruction **21**

4.1 Accumulation from Infinitesimal Influence 21

4.2 The Fundamental Theorem of Calculus 22

4.3 Change of Variables and Structural Invariance 22

5 Differential Equations and Dynamics 24

5.1 Rates as Laws of Motion 24

5.2 Equilibria and Linearisation 25

5.3 Systems of Differential Equations 25

6 Curvature, Constraint, and Manifolds 27

6.1 Motion Under Constraint 27

6.2 Curvature and Geodesics 27

6.3 The Structural Unity of Calculus 28

7 Multivariable Integration Theory 29

7.1 Measure and Volume in Higher Dimensions 29

7.2 Iterated Integrals and Fubini's Theorem 29

7.3 Coordinate Transformations and the Jacobian Determinant 30

7.4 Integration over Curves and Surfaces 30

7.5 Orientation and Differential Forms 30

8 Vector Calculus and the Geometry of Fields 31

8.1 Gradient, Divergence, and Curl 31

8.2 The Divergence Theorem and Stokes' Theorem 32

9 Nonlinear Dynamics and Phase Space 33

9.1 Phase Portraits and Flow Geometry 33

9.2 The Hartman–Grobman Theorem 34

9.3 Lyapunov Functions and Energy Methods 34

9.4 Bifurcation Theory 34

10 Constrained Dynamics and Lagrangian Structure **36**

10.1 Variational Principles and the Euler–Lagrange Equation 36

10.2 Hamiltonian Systems 37

10.3 Noether's Theorem 37

11 Geometric Control Theory **38**

11.1 Control Systems as Vector Fields with Input 38

11.2 Controllability and Lie Brackets 38

11.3 Optimal Control and the Pontryagin Maximum Principle 39

11.4 Control as Directed Tangent Dynamics 39

12 Structural Synthesis **41**

12.1 The Geometric Unification 41

12.2 Integration as Global Reconstruction from Local Flow 42

12.3 Stability as Curvature Constraint 42

13 From Differential Equations to Programs **43**

13.1 State Mutation Versus Functional Evolution 43

13.2 Time Discretisation as Approximation of Flow 43

13.3 Numerical Stability as Spectral Alignment 44

13.4 From Mutation to Morphism 44

14 Critique of Imperative Physiological Simulation **46**

14.1 The Architecture of Table-Driven Systems 46

14.2 Order Dependence and Hidden Coupling 46

14.3 Implicit Jacobians and Structural Opacity 47

14.4 The Structural Alternative 47

15 Calculus as a Functional Programming Language 49

15.1 Compositionality 49

15.2 Flows as Exponentials of Vector Fields 50

15.3 Automatic Differentiation 50

15.4 Calculus as a Declarative Language 50

16 Symbolic Versus Numerical Modelling 52

16.1 Structural Invariants in Symbolic Form 52

16.2 Truncation Error and Invariant Drift 52

16.3 Conservation-Preserving Methods 53

16.4 Modelling as Layered Representation 53

17 Retinal Cell as Dynamical System 54

17.1 Membrane Potential as a State Variable 54

17.2 Gating Variables as Coupled Dynamics 54

17.3 Phototransduction as Nested Dynamics 55

17.4 Functional Decomposition 55

17.5 Linearisation and Stability 56

18 Skin Cell and Boundary Regulation 57

18.1 Epidermal Dynamics as Reaction–Diffusion System 57

18.2 Boundary Conditions as Structural Constraints 57

18.3 Turing Instability and Pattern Formation 58

18.4 Fibrosis as Bifurcation in Repair Dynamics 58

19 Appliances as Controlled Dynamical Systems **59**

19.1 Thermal System: The Oven Model 59

19.2 Washing Machine Drum 60

19.3 Compositional Architecture 60

20 A Functional Simulation Architecture **61**

20.1 State as a Structured Object 61

20.2 Pure Vector Field Definition 61

20.3 Integrator as Separate Layer 62

20.4 Constraint Projection as Operator 62

20.5 Alignment with Calculus 63

21 Worked Example I: Minimal Retinal Cell **64**

21.1 The FitzHugh–Nagumo Model 64

21.2 Jacobian and Local Sensitivity 64

21.3 Numerical Simulation 65

21.4 Control Interpretation 65

22 Worked Example II: Skin Cell and Boundary Repair **66**

22.1 Minimal Repair Model 66

22.2 Phase Plane and Equilibrium Analysis 67

22.3 Projection Enforcement 67

23 Manifold-Aligned Inference and the Discipline of Not Predicting Noise **68**

23.1 From Dynamical Systems to Inference 68

23.2 Signal and Noise as Geometric Decomposition 68

23.3 Curvature Governs Generalisation 69

23.4 Noise Characterisation 69

23.5 Regularisation as Geometric Constraint 70

24 Entropy, Information, and Curvature in Model Space 71

24.1 The Fisher Information Metric 71

24.2 Free Energy as Variational Functional 71

24.3 Stability Requires Curvature–Entropy Alignment 72

25 The Unified Geometry of Physiology, Engineering, and Inference 73

25.1 One Architecture, Three Domains 73

25.2 A Single Geometric Doctrine 73

26 Models, Manifolds, and Adequacy 75

26.1 What a Model Is 75

26.2 Dimensionality and the Curse 75

26.3 Limits of Fixed-Phase-Space Modelling 75

27 Ergodicity and the Limits of Sampling 77

27.1 The Ergodic Theorem 77

27.2 Non-Ergodicity and Phase-Space Fragmentation 77

27.3 Stationarity and Distributional Drift 78

28 The Absolute Limits of Approximation 79

28.1 Approximation as Projection onto a Hypothesis Manifold 79

28.2 Five Geometric Limits 79

28.3 Where Curvature Is Stable, Refinement Succeeds 80

Chapter 1

Change Before Computation

1.1 The Problem of Variation

Calculus arises from a single question: how does one quantity change in response to another? This question is older than formal mathematics. It appears whenever motion is observed, whenever growth is measured, whenever tension balances across a structure, or whenever error is reduced through refinement.

The essential feature of change is *relational*. A quantity does not change in isolation; it changes *with respect to something*. Distance changes with respect to time. Pressure changes with respect to volume. Profit changes with respect to production. A curve bends because neighbouring values fail to align linearly.

Before numbers, before symbols, there is deformation. A line tilts. A surface bends. A system responds to perturbation. Calculus studies these responses.

Key Insight

Calculus is the geometry of relationship. Its central objects—derivatives, integrals, differential equations—encode how structured systems deform under infinitesimal perturbation and how that deformation accumulates into global form.

1.2 Rise, Run, and the Balance of Error

Consider a straight wall. Its slope is defined by rise over run. One may ascend a large vertical distance while advancing only slightly horizontally, and the ratio of these displacements determines steepness. Crucially, this ratio does not depend on absolute scale but on *proportional change*.

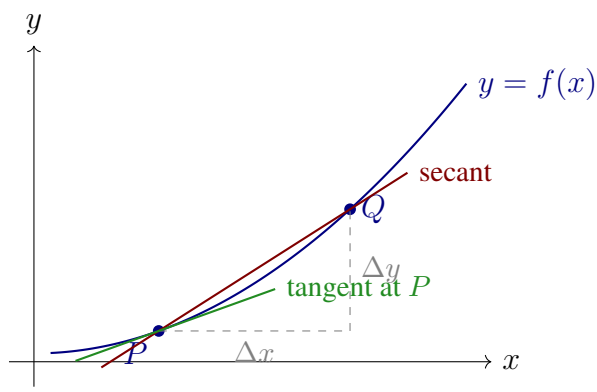


Figure 1.1: As Q approaches P along the curve, the secant line (Maroon) rotates toward the tangent line (green). The limiting direction is the derivative.

Now consider not a wall but a curve. A curve does not have a single slope; its steepness varies from point to point. To measure its inclination at a single location, we approximate it by secant lines connecting nearby points. As those points approach one another, the secant line approaches a unique limiting line: the tangent (fig. 1.1).

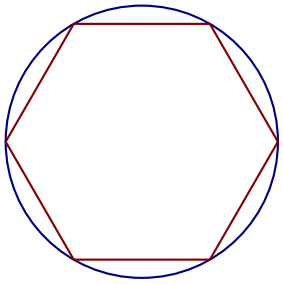
The tangent line balances error symmetrically. It is the line for which deviations on one side cancel those on the other to first order. This balancing process resembles a lever in equilibrium: two equal perturbations placed symmetrically about a point must balance to produce equilibrium.

Thus the derivative, at its core, is a *local balance condition*. It is the unique linear relation that stabilises infinitesimal variation.

1.3 Limits as Stability Under Refinement

The concept of limit formalises the idea of refinement. Suppose one estimates the circumference of a circle by inscribing polygons with increasingly many sides. As the number of sides increases, the perimeter approaches a fixed value. One approaches the

same number from above and below.



Hexagon ($n = 6$): perimeter $\to \pi d$ as $n \to \infty$

Figure 1.2: Inscribed polygons approximate the circle's circumference. Convergence is stability under refinement.

Convergence is not merely approximation—it is *agreement under arbitrarily small perturbation*.

Definition 1.1 (Limit)

A function $f(x)$ approaches L as $x \to a$, written $\lim_{x \to a} f(x) = L$, if for every tolerance $\varepsilon > 0$ there exists $\delta > 0$ such that

$$0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon.$$

This definition encodes stability. No matter how tightly one constrains acceptable error, one can restrict attention sufficiently to guarantee compliance. The limit is the expression of *controlled variation*.

Example

Let $f(x) = \frac{x^2-1}{x-1}$ for $x \neq 1$. Although f is not defined at $x = 1$, we compute $\lim_{x \to 1} \frac{x^2-1}{x-1} = \lim_{x \to 1}(x + 1) = 2$. The limit captures the behaviour of f near 1 without requiring evaluation at 1.

1.4 The Derivative as a Sensitivity Operator

Definition 1.2 (Derivative)

The derivative of $f : \mathbb{R} \to \mathbb{R}$ at x is

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h},$$

provided this limit exists.

This expression measures how much f responds to a small perturbation in x. The numerator represents output change; the denominator represents input change; their ratio is *relational sensitivity*.

When the limit exists, f is locally approximable by a linear map:

$$f(x+h) = f(x) + f'(x)h + o(h).$$

The notation $o(h)$ means a term that vanishes faster than h as $h \to 0$.

Key Insight

The derivative is not merely a number. It is the coefficient of the best local linear model—the linear approximation that most faithfully captures the function's behaviour near a point.

Constants vanish under differentiation because they exhibit no relational sensitivity. Linear terms become constants because their deformation is uniform. Higher-degree terms reduce degree because curvature itself varies. The derivative strips away inert components and retains only those parts that actively participate in change.

Example Power rule

For $f(x) = x^n$ with $n \in \mathbb{Z}_{>0}$, the binomial theorem gives

$$\frac{(x+h)^n - x^n}{h} = nx^{n-1} + \binom{n}{2} x^{n-2} h + \cdots \xrightarrow{h \to 0} nx^{n-1}.$$

Thus $(x^n)' = nx^{n-1}$, the power rule.

1.5 Abstraction as Reduction of Irrelevant Degrees of Freedom

When differentiating

$$f(x) = ax^3 + bx^2 + cx + d,$$

the result

$$f'(x) = 3ax^2 + 2bx + c$$

reveals structure independent of arithmetic complication. The transformation rule does not depend on the magnitude of a, b, c, d. It depends only on exponentiation and linearity. Abstraction removes irrelevant degrees of freedom while preserving relational form.

This principle recurs throughout the text. Calculus is not a collection of tricks but a disciplined method for isolating structural relationships from computational noise.

Chapter 2

Higher-Order Change and Curvature

2.1 The Second Derivative and the Geometry of Bending

The second derivative measures the rate at which the slope itself changes:

$$f''(x) = \frac{d}{dx}\big(f'(x)\big).$$

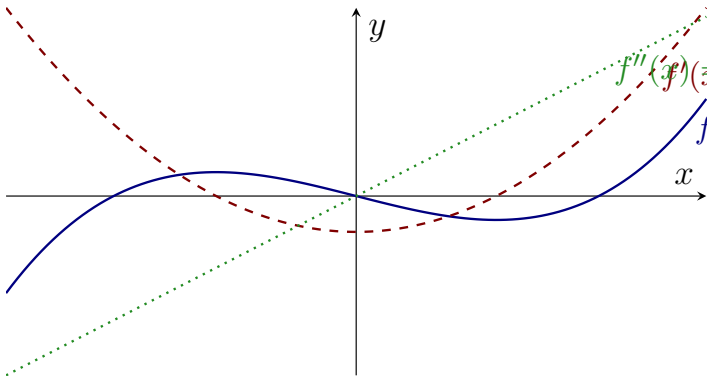


Figure 2.1: A function (blue), its first derivative (red dashed), and second derivative (green dotted). Where $f'' > 0$ the function is concave up; where $f'' < 0$ it is concave down.

The Taylor expansion makes the role of the second derivative explicit:

$$f(x+h) = f(x) + f'(x)h + \frac{1}{2}f''(x)h^2 + o(h^2).$$

The quadratic term encodes bending. It quantifies the deviation from perfect linearity.

Higher derivatives describe successive layers of structural refinement: each derivative reveals how deformation itself deforms.

2.2 Critical Points and Stability

Definition 2.1 (Critical point)

A point x_0 is *critical* for f if $f'(x_0) = 0$.

Near a critical point,

$$f(x_0 + h) = f(x_0) + \tfrac{1}{2} f''(x_0) h^2 + o(h^2).$$

Since $h^2 \geq 0$, the sign of $f''(x_0)$ determines whether perturbations raise or lower the function.

Theorem 2.1 Second Derivative Test

Let $f'(x_0) = 0$.

- If $f''(x_0) > 0$, then x_0 is a local minimum.
- If $f''(x_0) < 0$, then x_0 is a local maximum.
- If $f''(x_0) = 0$, the test is inconclusive.

Key Insight

Stability is a curvature condition. Positive curvature confines nearby trajectories; negative curvature repels them.

2.3 From One Variable to Many

Real systems rarely depend on a single variable. For $f : \mathbb{R}^n \to \mathbb{R}$, each coordinate may vary independently.

Definition 2.2 (Partial derivative)

The partial derivative of f with respect to x_i is

$$\frac{\partial f}{\partial x_i}(x) = \lim_{h \to 0} \frac{f(x_1, \ldots, x_i + h, \ldots, x_n) - f(x)}{h}.$$

Collecting all partial derivatives yields the *gradient vector*:

$$\nabla f(x) = \begin{pmatrix} \partial f / \partial x_1 \\ \vdots \\ \partial f / \partial x_n \end{pmatrix}.$$

The gradient is not a collection of slopes but a *directional sensitivity field*.

2.4 Directional Derivatives and Linear Structure

Let $v \in \mathbb{R}^n$ be a unit direction vector. The directional derivative is

$$D_v f(x) = \lim_{h \to 0} \frac{f(x + hv) - f(x)}{h}.$$

When f is differentiable,

$$D_v f(x) = \nabla f(x) \cdot v.$$

The gradient therefore represents the *linear functional* that maps directional perturbations to scalar responses. Differentiability in $\mathbb{R}^n$ means that f admits a linear approximation:

$$f(x + h) = f(x) + \nabla f(x) \cdot h + o(\|h\|).$$

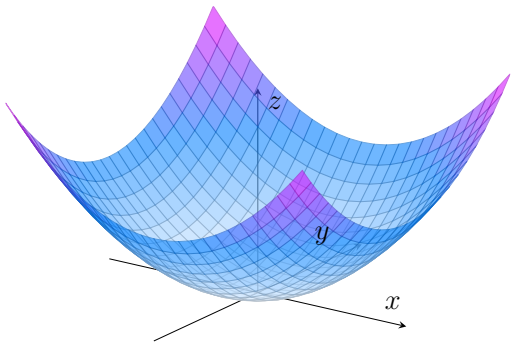


Figure 2.2: The paraboloid $f(x, y) = x^2 + y^2$. The gradient $\nabla f = (2x, 2y)$ points radially outward — in the direction of steepest ascent — and is always perpendicular to the level curves (circles).

Chapter 3

Tangent Spaces and Local Models

3.1 Curves and Embedded Surfaces

At a point p on a smooth surface $S \subset \mathbb{R}^3$, curves lying on S have velocity vectors at p. The collection of all such velocity vectors forms a plane: the tangent plane.

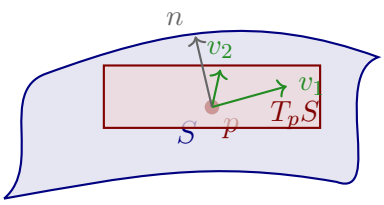


Figure 3.1: The tangent space T_pS (red plane) at a point p on a surface S (blue). Tangent vectors v_1, v_2 lie in the plane; the normal n is perpendicular to it.

Definition 3.1 (Tangent space of a manifold)

For a smooth manifold M and a point $x \in M$, the tangent space is

$$T_xM = \{\gamma'(0) \mid \gamma : (-\varepsilon, \varepsilon) \to M,\ \gamma(0) = x\}.$$

The tangent space captures all possible instantaneous directions of motion constrained to M.

3.2 Functions Between Manifolds and the Jacobian

Let $F : M \to N$ be a smooth map between manifolds. The derivative of F at x is the linear map

$$dF_x : T_x M \to T_{F(x)} N.$$

For $F : \mathbb{R}^n \to \mathbb{R}^m$, this linear map is represented by the Jacobian matrix:

$$J_F(x) = \begin{pmatrix} \frac{\partial F_1}{\partial x_1} & \cdots & \frac{\partial F_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial x_1} & \cdots & \frac{\partial F_m}{\partial x_n} \end{pmatrix}.$$

The Jacobian is the multivariable generalisation of slope.

3.3 The Chain Rule as Composition of Deformation

Theorem 3.1 Chain Rule

For smooth maps $F : M \to N$ and $G : N \to P$,

$$d(G \circ F)_x = dG_{F(x)} \circ dF_x.$$

The derivative of a composition is the composition of derivatives. In coordinates:

$$J_{G \circ F}(x) = J_G(F(x)) \cdot J_F(x).$$

Key Insight

The chain rule is not computational coincidence; it is a categorical necessity of linear approximation. Local deformation in M maps into N, then into P. Sensitivity composes.

$$\begin{tikzcd} T_xM \arrow[r, "dF_x"] \arrow[dr, "d(G\circ F)_x"'] & T_{F(x)}N \arrow[d, "dG_{F(x)}"] \\ & T_{(G\circ F)(x)}P \end{tikzcd}$$

Figure 3.2: The chain rule as commutative diagram: the composite derivative equals the composition of individual derivatives.

3.4 The Hessian and Second-Order Structure

For $f : \mathbb{R}^n \to \mathbb{R}$, the second derivative is encoded in the Hessian matrix:

$$H_f(x) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}\right)_{i,j}.$$

The quadratic Taylor expansion is

$$f(x+h) = f(x) + \nabla f(x) \cdot h + \tfrac{1}{2} h^\top H_f(x) h + o(\|h\|^2).$$

Theorem 3.2 Multivariable Second Derivative Test

Let $\nabla f(x_0) = 0$. Stability at x_0 is determined by the eigenvalues of $H_f(x_0)$:

- All eigenvalues positive $\Rightarrow$ local minimum.
- All eigenvalues negative $\Rightarrow$ local maximum.
- Mixed signs $\Rightarrow$ saddle point.

Chapter 4

Integration and Reconstruction

4.1 Accumulation from Infinitesimal Influence

If the derivative measures how a quantity changes locally, the integral measures how local changes accumulate into global structure. The two operations are inverse procedures within a unified stability framework.

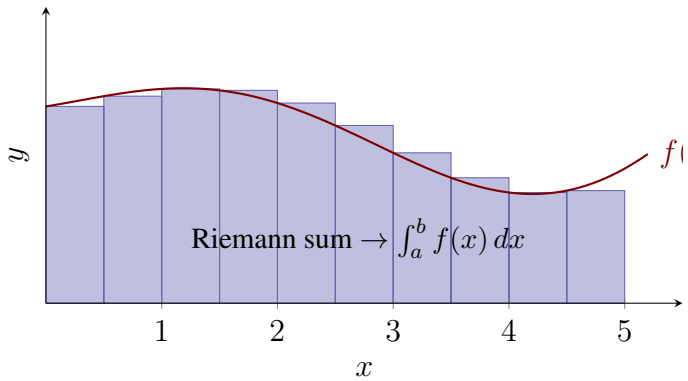


Figure 4.1: A Riemann sum (blue rectangles) approximating the integral of f over $[0, 5]$. As rectangle widths shrink to zero, the sum converges to the definite integral.

Definition 4.1 (Definite integral)

If $f : [a, b] \to \mathbb{R}$ and the Riemann sums

$$S_n = \sum_{i=0}^{n-1} f(x_i^*)\,\Delta x_i$$

converge as $\max_i \Delta x_i \to 0$ (independently of the choice of sample points), the limit is the *definite integral*

$$\int_a^b f(x)\,dx.$$

4.2 The Fundamental Theorem of Calculus

Theorem 4.1 Fundamental Theorem of Calculus

Let f be continuous on $[a, b]$.

(i) If $F(x) = \int_a^x f(t)\,dt$, then $F'(x) = f(x)$.

(ii) If $F'(x) = f(x)$, then $\int_a^b f(x)\,dx = F(b) - F(a)$.

Key Insight

The Fundamental Theorem is not merely computational convenience. It expresses that global structure can be reconstructed from local sensitivity, and local sensitivity can be extracted from global structure. The integral aggregates deformation; the derivative isolates it.

Proof sketch of Part (i). By definition, $F'(x) = \lim_{h\to 0} \frac{1}{h}\int_x^{x+h} f(t)\,dt$. By continuity of f, for any $\varepsilon > 0$ there exists $\delta > 0$ such that $|f(t) - f(x)| < \varepsilon$ whenever $|t - x| < \delta$. Then

$$\left|\frac{1}{h}\int_x^{x+h} f(t)\,dt - f(x)\right| = \left|\frac{1}{h}\int_x^{x+h} (f(t) - f(x))\,dt\right| \le \varepsilon.$$

Since ε was arbitrary, $F'(x) = f(x)$. ■

4.3 Change of Variables and Structural Invariance

For a transformation $x = g(u)$:

$$\int f(x)\,dx = \int f(g(u))\,g'(u)\,du.$$

In multiple dimensions, if $\Phi : \mathbb{R}^n \to \mathbb{R}^n$ is smooth with Jacobian determinant $\det J_\Phi$,

then

$$\int_{\Phi(U)} h(x)\,dx = \int_U h(\Phi(u))\,|\det J_\Phi(u)|\,du.$$

The Jacobian determinant measures how *volume scales* under the transformation. Integration must incorporate this geometric correction to remain invariant.

Example Polar coordinates

The transformation $x = r\cos\theta$, $y = r\sin\theta$ has $\det J = r$, giving the familiar formula $\iint_\Omega f(x,y)\,dx\,dy = \int_0^{2\pi}\int_0^R f(r\cos\theta, r\sin\theta)\,r\,dr\,d\theta$.

Chapter 5

Differential Equations and Dynamics

5.1 Rates as Laws of Motion

A differential equation prescribes how a quantity evolves in response to its current state. It defines a rule of motion rather than a static relationship.

Definition 5.1 (Ordinary differential equation)

A first-order ODE is an equation of the form

$$\frac{dx}{dt} = f(x, t),$$

where $x : \mathbb{R} \to \mathbb{R}$ (or $\mathbb{R}^n$) is unknown and f specifies instantaneous rate of change.

Solving a differential equation means *reconstructing global behaviour from a local law*.

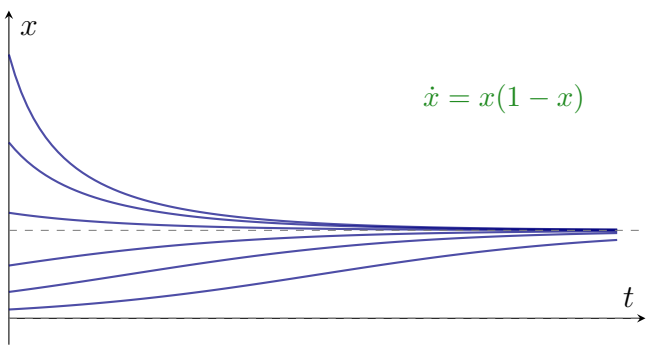


Figure 5.1: Phase portrait for the logistic equation $\dot{x} = x(1-x)$. All trajectories with $x(0) > 0$ converge to $x = 1$ (stable equilibrium); $x = 0$ is unstable.

5.2 Equilibria and Linearisation

An equilibrium x_0 satisfies $f(x_0) = 0$. Near equilibrium, a Taylor expansion gives the *linearised equation*:

$$\frac{d}{dt}(x - x_0) = f'(x_0)(x - x_0) + \text{higher-order terms.}$$

Theorem 5.1 Stability via linearisation

Let x_0 be an equilibrium of $\dot{x} = f(x)$.

- If $f'(x_0) < 0$, perturbations decay exponentially; x_0 is *asymptotically stable*.
- If $f'(x_0) > 0$, perturbations grow; x_0 is *unstable*.

In higher dimensions, stability depends on the eigenvalues of $J_f(x_0)$.

5.3 Systems of Differential Equations

For $x \in \mathbb{R}^n$, the system $\dot{x} = F(x)$ defines a *vector field* on state space. Each initial condition generates a trajectory determined by the integral curves of F.

For linear systems $\dot{x} = Ax$, the solution is

$$x(t) = e^{At}x(0),$$

where e^{At} is the matrix exponential. The eigenvalues of A determine long-term behaviour:

- Negative real parts $\Rightarrow$ exponential decay.
- Positive real parts $\Rightarrow$ exponential growth.
- Purely imaginary parts $\Rightarrow$ oscillation.

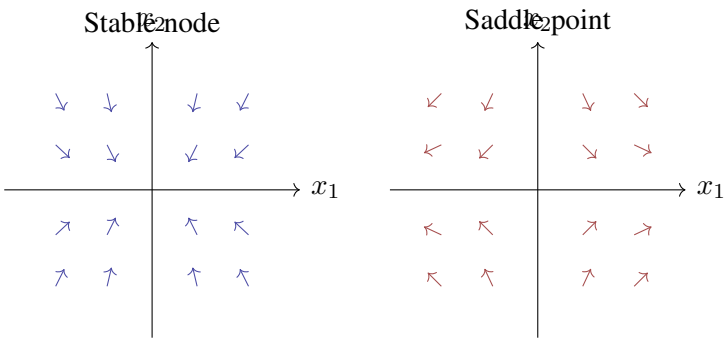


Figure 5.2: Phase portraits for a stable node (left, both eigenvalues negative) and a saddle point (right, eigenvalues of opposite sign).

Chapter 6

Curvature, Constraint, and Manifolds

6.1 Motion Under Constraint

If motion is restricted to a manifold M, the differential equation becomes

$$\frac{dx}{dt} = \Pi_{T_x M} F(x),$$

where $\Pi_{T_x M}$ denotes orthogonal projection onto the tangent space. This ensures trajectories remain on the manifold by suppressing normal components (fig. 3.1).

Example Pendulum

A pendulum of length ℓ is constrained to the circle of radius ℓ. The unconstrained gravity vector is projected onto the tangential direction, yielding $\ddot{\theta} = -(g/\ell)\sin\theta$.

6.2 Curvature and Geodesics

On a Riemannian manifold (M, g), geodesics are curves that locally minimise length. They satisfy the geodesic equation

$$\nabla_{\dot{\gamma}} \dot{\gamma} = 0,$$

where ∇ is the Levi-Civita connection.

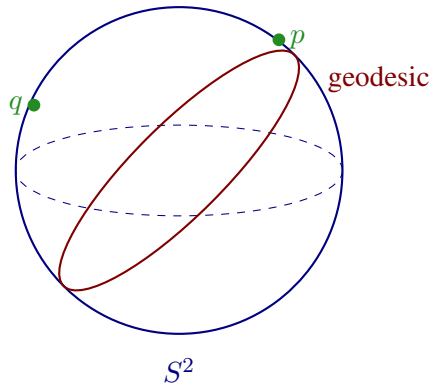


Figure 6.1: A geodesic (great circle) on the sphere S^2 connecting points p and q. The geodesic minimises length; curvature causes geodesics to converge at poles.

The *Riemann curvature tensor* R measures how geodesics initially parallel deviate from one another under parallel transport. Positive curvature (as on a sphere) causes convergence; negative curvature (as on a hyperbolic surface) causes divergence.

6.3 The Structural Unity of Calculus

The themes developed in the first six chapters can be summarised in a single diagram:

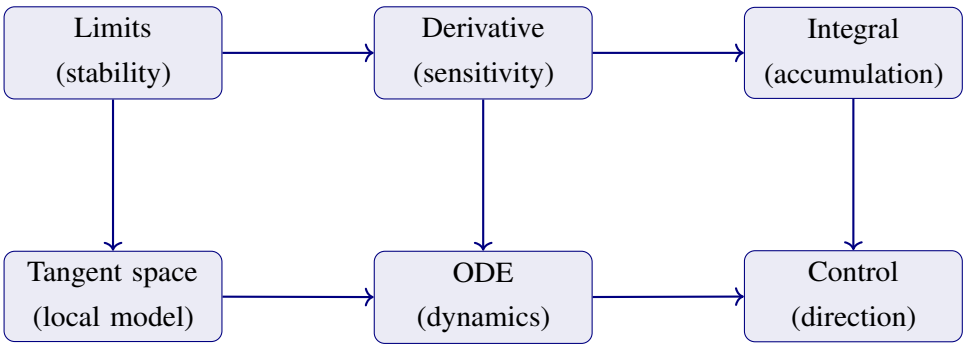


Figure 6.2: Structural dependencies among the core concepts of calculus.

Chapter 7

Multivariable Integration Theory

7.1 Measure and Volume in Higher Dimensions

For a bounded region $\Omega \subset \mathbb{R}^n$, the integral of $f : \Omega \to \mathbb{R}$ is defined by partitioning into small subregions of volume ΔV_i:

$$\int_\Omega f(x)\,dx = \lim_{\max \Delta V_i \to 0} \sum_i f(x_i^*)\,\Delta V_i.$$

7.2 Iterated Integrals and Fubini's Theorem

Theorem 7.1 Fubini

If $f : \Omega \to \mathbb{R}$ is integrable on $\Omega = [a, b] \times [c, d]$, then

$$\int_\Omega f(x, y)\,dx\,dy = \int_a^b \int_c^d f(x, y)\,dy\,dx = \int_c^d \int_a^b f(x, y)\,dx\,dy.$$

The significance is structural: accumulation along independent coordinate directions is compatible with accumulation over the whole region. Volume measure factorises across orthogonal directions.

7.3 Coordinate Transformations and the Jacobian Determinant

For a smooth bijection $\Phi : U \to \Omega$,

$$\int_\Omega f(x)\,dx = \int_U f(\Phi(u))\,|\det D\Phi(u)|\,du.$$

The Jacobian determinant measures local volume distortion. It is the net scaling factor of the linear map $D\Phi(u)$, capturing how infinitesimal cubes deform under the transformation.

7.4 Integration over Curves and Surfaces

For a smooth curve $\gamma : [a, b] \to \mathbb{R}^n$, the arc-length integral is

$$\int_\gamma f\,ds = \int_a^b f(\gamma(t))\,\|\gamma'(t)\|\,dt.$$

For a surface S parameterised by $\Phi(u, v)$:

$$\int_S f\,dS = \int_U f(\Phi(u, v))\,\|\partial_u\Phi \times \partial_v\Phi\|\,du\,dv.$$

In both cases: integration equals *accumulation of infinitesimal geometric elements weighted by local density*.

7.5 Orientation and Differential Forms

A differential k-form assigns oriented infinitesimal k-dimensional volume elements to tangent spaces. Orientation determines the sign of accumulated quantities. The exterior derivative d maps k-forms to $(k + 1)$-forms and satisfies $d^2 = 0$, encoding the fact that *boundaries of boundaries vanish*.

Chapter 8

Vector Calculus and the Geometry of Fields

8.1 Gradient, Divergence, and Curl

Three differential operators play central roles in vector calculus.

Definition 8.1 (Divergence)

For $F : \mathbb{R}^3 \to \mathbb{R}^3$,

$$\nabla \cdot F = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}.$$

Divergence measures local volumetric expansion ($\nabla \cdot F > 0$) or compression ($\nabla \cdot F < 0$).

Definition 8.2 (Curl)

$$\nabla \times F = \begin{pmatrix} \partial_y F_3 - \partial_z F_2 \\ \partial_z F_1 - \partial_x F_3 \\ \partial_x F_2 - \partial_y F_1 \end{pmatrix}.$$

Curl measures local rotation density.

$\nabla \cdot F > 0$ (source) $\quad \nabla \times F \neq 0$ (rotation)

Figure 8.1: Left: a diverging field with positive divergence (source). Right: a rotating field with nonzero curl.

8.2 The Divergence Theorem and Stokes' Theorem

Theorem 8.1 Divergence Theorem

For a region Ω with outward-pointing boundary normal n,

$$\int_{\Omega} \nabla \cdot F \, dV = \int_{\partial\Omega} F \cdot n \, dS.$$

Total interior expansion equals boundary flux.

Theorem 8.2 Stokes' Theorem

For an oriented surface S with boundary ∂S,

$$\int_{S} (\nabla \times F) \cdot n \, dS = \int_{\partial S} F \cdot dr.$$

Total curl over a surface equals boundary circulation.

Both theorems are special cases of the general Stokes theorem for differential forms:

$$\int_{\partial M} \omega = \int_{M} d\omega.$$

Boundary behaviour reflects interior variation. This is the geometric expression of conservation laws.

Chapter 9

Nonlinear Dynamics and Phase Space

9.1 Phase Portraits and Flow Geometry

A nonlinear dynamical system

$$\frac{dx}{dt} = F(x), \qquad F : \mathbb{R}^n \to \mathbb{R}^n,$$

assigns to each state an instantaneous direction of evolution. The collection of all trajectories forms the *flow* of the system.

Van der Pol oscillator phase portrait

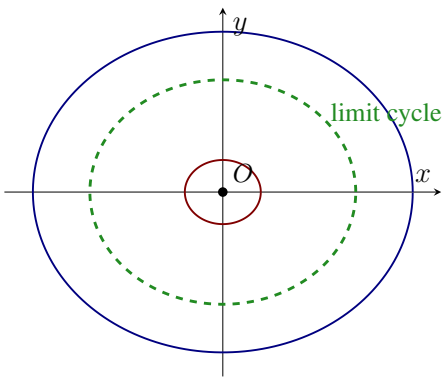


Figure 9.1: Schematic phase portrait of a limit cycle system. Trajectories starting far out (blue) spiral inward; trajectories starting near the origin (red) spiral outward. Both converge to the limit cycle (green dashed circle).

9.2 The Hartman–Grobman Theorem

Theorem 9.1 Hartman–Grobman

If x_0 is a hyperbolic equilibrium of $\dot{x} = F(x)$ (i.e. no eigenvalue of $DF(x_0)$ lies on the imaginary axis), then the nonlinear system near x_0 is *topologically conjugate* to its linearisation $\dot{y} = DF(x_0)\, y$.

Thus, near hyperbolic equilibria, nonlinear dynamics inherit qualitative behaviour from linear systems. The eigenvalues of $DF(x_0)$ give a complete local portrait.

9.3 Lyapunov Functions and Energy Methods

Definition 9.1 (Lyapunov function)

A smooth function $V : \mathbb{R}^n \to \mathbb{R}_{\geq 0}$ is a Lyapunov function for $\dot{x} = F(x)$ at equilibrium x_0 if $V(x_0) = 0$, $V(x) > 0$ for $x \neq x_0$ nearby, and $\dot{V}(x) := \nabla V(x) \cdot F(x) \leq 0$ along trajectories.

Theorem 9.2 Lyapunov stability

If a Lyapunov function V exists with $\dot{V} < 0$ (strict), then x_0 is asymptotically stable.

Lyapunov functions generalise potential functions to non-conservative systems. They provide stability certificates without requiring explicit solution of the differential equation.

9.4 Bifurcation Theory

As a parameter μ varies, equilibria may appear, disappear, or change stability. This is a *bifurcation*.

Example Pitchfork bifurcation

Consider $\dot{x} = \mu x - x^3$. For $\mu < 0$, the only equilibrium is $x = 0$ (stable). For $\mu > 0$, the origin becomes unstable and two new stable equilibria $x = \pm\sqrt{\mu}$ emerge (fig. 9.2).

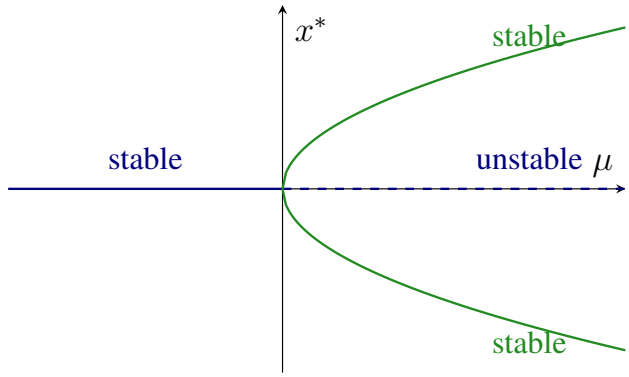


Figure 9.2: Bifurcation diagram for the pitchfork bifurcation $\dot{x} = \mu x - x^3$. At $\mu = 0$, one stable equilibrium splits into three.

Chapter 10

Constrained Dynamics and Lagrangian Structure

10.1 Variational Principles and the Euler–Lagrange Equation

The calculus of variations generalises optimisation from functions to *functionals*.

Definition 10.1 (Action functional)

Given a Lagrangian $L(q, \dot{q}, t)$, the action is

$$\mathcal{S}[\gamma] = \int_a^b L(\gamma(t), \dot{\gamma}(t), t)\, dt.$$

Theorem 10.1 Euler–Lagrange equation

Critical curves of $\mathcal{S}$ satisfy

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\gamma}} - \frac{\partial L}{\partial \gamma} = 0.$$

This equation arises by demanding that the first-order variation of the action vanishes under all compactly supported perturbations. Motion becomes a *stationary condition* in function space.

Key Insight

The Euler–Lagrange equation expresses dynamics as extremisation of accumulated quantities. Geodesics, classical trajectories, and elastica are all solutions to variational problems.

10.2 Hamiltonian Systems

Define conjugate momentum $p = \partial L / \partial \dot{q}$ and Hamiltonian

$$H(q, p) = p \cdot \dot{q} - L.$$

Theorem 10.2 Hamilton's equations

$$\dot{q} = \frac{\partial H}{\partial p}, \qquad \dot{p} = -\frac{\partial H}{\partial q}.$$

Hamiltonian flows preserve the symplectic form $\omega = dp \wedge dq$ and, when H is time-independent, conserve energy. Phase space volume is also conserved (Liouville's theorem).

10.3 Noether's Theorem

Theorem 10.3 Noether

If the action $\mathcal{S}$ is invariant under a one-parameter family of transformations generated by a vector field X, then the quantity

$$Q = \frac{\partial L}{\partial \dot{\gamma}} \cdot X(\gamma)$$

is conserved along solutions.

Symmetry generates conservation. Temporal invariance gives energy conservation; spatial translational invariance gives momentum conservation; rotational invariance gives angular momentum conservation.

Chapter 11

Geometric Control Theory

11.1 Control Systems as Vector Fields with Input

A control system extends a dynamical system by introducing external inputs:

$$\frac{dx}{dt} = F(x) + \sum_{i=1}^{m} u_i \, G_i(x),$$

where F is the drift field, G_i are control vector fields, and $u_i(t)$ are chosen inputs. Control introduces *directed deformation*.

11.2 Controllability and Lie Brackets

Definition 11.1 (Lie bracket)

For smooth vector fields X, Y on $\mathbb{R}^n$, the Lie bracket is

$$[X, Y](x) = DY(x)\, X(x) - DX(x)\, Y(x).$$

Lie brackets represent second-order motion generated by alternating flows. Physically, flowing along X for time $\sqrt{h}$, then Y, then $-X$, then $-Y$ produces a displacement of order h in direction $[X, Y]$.

Theorem 11.1 Chow–Rashevskii (accessibility)

If the distribution generated by $\{G_1, \ldots, G_m\}$ and all their iterated Lie brackets spans $T_x\mathbb{R}^n$ at every x, then the system is *accessible* (reachable from any point to any other).

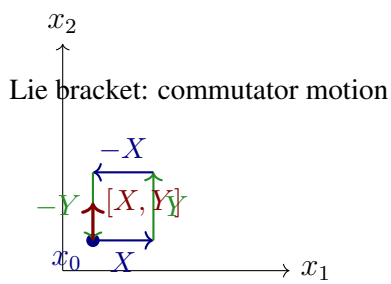


Figure 11.1: Geometric interpretation of the Lie bracket. Alternating unit flows along X, Y, $-X$, $-Y$ over time $\sqrt{h}$ produce a net displacement of order h in the direction $[X, Y]$.

11.3 Optimal Control and the Pontryagin Maximum Principle

Optimal control minimises a cost functional

$$J(u) = \int_0^T L(x(t), u(t))\, dt.$$

The *Pontryagin Maximum Principle* introduces adjoint (costate) variables $p(t)$ and a Hamiltonian

$$\mathcal{H}(x, p, u) = p \cdot F(x, u) - L(x, u).$$

Optimal trajectories satisfy Hamilton's equations for $\mathcal{H}$ together with the maximality condition $u^*(t) = \arg\max_u \mathcal{H}(x, p, u)$.

11.4 Control as Directed Tangent Dynamics

Throughout calculus we have studied local linearisation, curvature, accumulation, and dynamical evolution. Control theory adds *intention*.

Where dynamics describe what *will* occur, control prescribes what *should* occur. The derivative measures sensitivity; the integral reconstructs trajectory; geometry constrains motion; control directs permissible variation toward desired equilibria.

Chapter 12

Structural Synthesis

12.1 The Geometric Unification

Across all preceding chapters, a single theme persists: calculus is the geometry of controlled change. A differentiable function is one that admits linear approximation. A dynamical system is one whose evolution is governed by local vector fields. A constrained system evolves within tangent spaces. A controlled system modifies those tangent directions intentionally.

Each structure rests upon the same foundation: *infinitesimal deformation aligned with geometric constraint*.

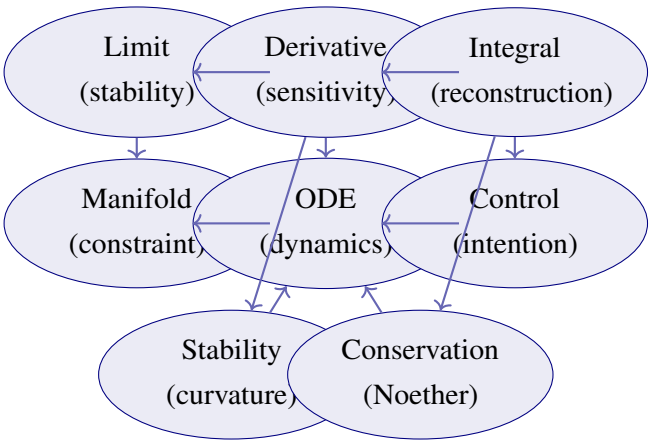


Figure 12.1: Conceptual map of calculus. Every arrow represents a logical or structural dependency developed in the text.

12.2 Integration as Global Reconstruction from Local Flow

The integral is not merely an area computation—it is the recovery of coherent structure from local contributions. In scalar calculus, the integral reconstructs accumulated quantity from instantaneous rate. In vector calculus, integral theorems reveal how boundary behaviour reflects interior variation. In variational calculus, the integral defines energy landscapes whose critical points determine motion.

12.3 Stability as Curvature Constraint

Equilibrium is governed by curvature. Second derivatives classify local extrema. Hessians encode multi-directional bending. Lyapunov functions capture energy descent. Eigenvalues determine asymptotic behaviour. On manifolds, curvature shapes geodesic deviation. In nonlinear systems, phase-space curvature governs trajectory divergence.

Stability is geometric discipline imposed by curvature.

Chapter 13

From Differential Equations to Programs

13.1 State Mutation Versus Functional Evolution

Many computational systems implement dynamics via state mutation:

$$x \leftarrow x + \Delta x.$$

The symbol $\leftarrow$ indicates replacement. From the perspective of calculus, however, this representation is incomplete.

A differential equation $\dot{x} = F(x)$ defines a *flow*: given initial state x_0, it determines a trajectory $x(t) = \Phi_t(x_0)$, where Φ_t is the flow map satisfying the *semigroup property*:

$$\Phi_{t+s} = \Phi_t \circ \Phi_s.$$

This expresses time consistency: evolving for time s then t equals evolving for $t + s$ directly. Mutation updates a variable; a flow defines a *transformation family*.

13.2 Time Discretisation as Approximation of Flow

Euler's method approximates the flow map at step h:

$$x_{n+1} = x_n + h\,F(x_n) \approx \Phi_h(x_n).$$

By Taylor expansion:

$$x(t+h) = x(t) + h\,F(x(t)) + \mathcal{O}(h^2).$$

Euler discards the $\mathcal{O}(h^2)$ terms. Higher-order schemes (Runge–Kutta) improve the approximation. Each scheme constructs an approximate flow operator $\tilde{\Phi}_h$ satisfying

$$\tilde{\Phi}_h(x) = \Phi_h(x) + \mathcal{O}(h^{p+1})$$

for some order p.

Key Insight

Discrete simulation is meaningful only insofar as it approximates a continuous geometric object. The physical model is the vector field F; the numerical scheme approximates its integral curves.

13.3 Numerical Stability as Spectral Alignment

For a linear system $\dot{x} = Ax$, the exact solution is $x(t) = e^{At}x(0)$. Euler's method produces $x_{n+1} = (I + hA)x_n$, so eigenvalues of the true dynamics $\lambda_i(A)$ map to eigenvalues $1 + h\lambda_i(A)$ of the update operator.

Stability requires $|1 + h\lambda_i| < 1$, imposing a step-size restriction proportional to the spectral radius of A.

Key Insight

Numerical stability is a spectral alignment problem. The eigenvalues of the numerical update operator must mirror the eigenvalues of the true flow.

13.4 From Mutation to Morphism

A mutation-based simulation hides geometry inside procedural detail. A flow-based simulation exposes geometry explicitly. The natural architectural principle is: represent dynamics as morphisms between state spaces rather than as sequences of overwrites.

If $F : X \to TX$ assigns tangent vectors, then integration constructs $\Phi_t : X \to$

X. Programs respecting this structure are compositional and admit differentiation, linearisation, stability analysis, and constraint projection naturally.

Chapter 14

Critique of Imperative Physiological Simulation

14.1 The Architecture of Table-Driven Systems

Large physiological simulators are often constructed using imperative paradigms. State variables are stored in tables; each time step executes a sequence of update procedures. A typical fragment takes the form:

$$V \leftarrow V + h \cdot f(V, m, n, \dots), \qquad m \leftarrow m + h \cdot g(V, m, \dots).$$

In such systems, state variables are globally mutable, update order is explicitly encoded, dependencies are implicit in code structure, and coupling is distributed across procedural fragments. The geometric object F exists implicitly but is never explicit.

14.2 Order Dependence and Hidden Coupling

Consider two coupled variables governed by

$$\dot{x} = f(x, y), \qquad \dot{y} = g(x, y).$$

A *simultaneous* Euler update evaluates both increments at (x_n, y_n). But sequential mutation—updating x first—evaluates g at (x_{n+1}, y_n) rather than (x_n, y_n). This introduces

an unintentional semi-implicit scheme without deliberate design.

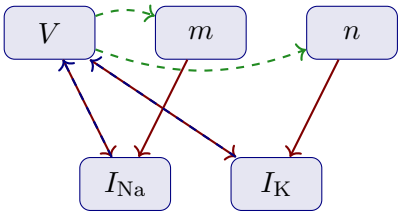
Remark

The vector field $F(x, y)$ defines a simultaneous infinitesimal direction in the tangent space of $\mathbb{R}^2$. Sequential mutation replaces geometric simultaneity with procedural causality.

14.3 Implicit Jacobians and Structural Opacity

In an imperative simulator, every update rule implicitly defines partial derivatives—but these are distributed across code. Without an explicit representation of F as a function from state space to tangent space, the Jacobian must be reconstructed by tracing procedural dependencies.

When F is explicit, $J_F(x) = DF(x)$ is immediate. Linearisation, eigenvalue computation, and stability classification follow directly.



Dependency graph (implicit Jacobian)

Figure 14.1: Even in a simple Hodgkin–Huxley fragment, the Jacobian is a directed graph embedded implicitly in code. Making F explicit renders the full matrix J_F directly computable.

14.4 The Structural Alternative

The critique is not that imperative systems are incorrect—many produce accurate results. The critique is structural: they *obscure the geometric object they approximate*.

When F is explicit:

- Linearisation is direct: $J_F(x) = DF(x)$.
- Stability analysis is systematic: inspect eigenvalues.

- Conservation laws are verifiable: check $\nabla I \cdot F = 0$.
- Constraint projection is implementable: $\Pi_{T_x M} F(x)$.
- Composition of subsystems is functional: $F = F_1 + F_2 + \cdots$.

Chapter 15

Calculus as a Functional Programming Language

15.1 Compositionality

Differentiation assigns to each smooth map $f : X \to Y$ a linear map between tangent spaces $Df_x : T_xX \to T_{f(x)}Y$. Crucially, it respects composition:

$$D(g \circ f)_x = Dg_{f(x)} \circ Df_x.$$

This means differentiation is *functorial*: it defines a functor $D : \mathbf{Diff} \to \mathbf{Lin}$, mapping the category of smooth manifolds with differentiable maps to the category of vector spaces with linear maps.

$$\begin{tikzcd}
(X, f) \arrow[r, "g \circ f"] \arrow[d, "D"] & (Z, \cdot) \arrow[d, "D"] \\
(T_xX, Df_x) \arrow[r, "D(g \circ f)_x"] & (T_{(g \circ f)(x)}Z, \cdot)
\end{tikzcd}$$

Figure 15.1: Functoriality of differentiation: the derivative of a composition equals the composition of derivatives.

15.2 Flows as Exponentials of Vector Fields

Given $F : X \to TX$, integration produces $\Phi_t : X \to X$. One writes formally

$$\Phi_t = \exp(tF),$$

the time-t map of $\dot{x} = F(x)$. Vector fields are *infinitesimal programs*; flows are their execution.

15.3 Automatic Differentiation

If a system is defined compositionally as $F = F_1 + F_2 \circ G$, derivatives propagate automatically via the chain rule. Automatic differentiation (AD) systems exploit this structure:

- *Forward mode*: propagate tangent vectors alongside evaluation.
- *Reverse mode*: propagate cotangent vectors (adjoints) backward.

The correctness of AD rests entirely on $D(g \circ f) = Dg \circ Df$. AD computes exact derivatives—up to floating-point precision—by structural recursion on expression graphs.

15.4 Calculus as a Declarative Language

A declarative program specifies *what* relationship holds, not *how* to mutate state step by step. The equation $\dot{x} = F(x)$ declares a relational deformation. It specifies no loop structure, no variable overwriting, and no evaluation order.

Calculus concept	**Programming analogue**	**Mechanism**
State manifold	Type	Structural constraint
Vector field $F : X \to TX$	Pure function	Referential transparency
Flow Φ_t	Fold over time	Functional iteration
Chain rule	Automatic differentiation	Compositional recursion
Projection Π_{TM}	Constraint enforcement	Type-level invariant
Jacobian DF	Sensitivity analysis	Derivative of pure function

Key Insight

Calculus, properly understood, is not merely a toolkit for computation. It is a language of compositional transformation. It is functional at its core.

Chapter 16

Symbolic Versus Numerical Modelling

16.1 Structural Invariants in Symbolic Form

When a dynamical system is written symbolically as $\dot{x} = F(x)$, its structure is transparent. A scalar function $I : X \to \mathbb{R}$ is an invariant if $\nabla I \cdot F = 0$, which can be verified algebraically.

Symbolic modelling preserves this transparency. The invariant is not an empirical observation—it is a structural property.

16.2 Truncation Error and Invariant Drift

Under Euler's method, an exact invariant I satisfies

$$I(x_{n+1}) = I(x_n) + \mathcal{O}(h^2).$$

Over $N \sim 1/h$ steps, the invariant drifts by $\mathcal{O}(h)$. This is not merely a numerical nuisance; it reflects a real departure from the constraint manifold.

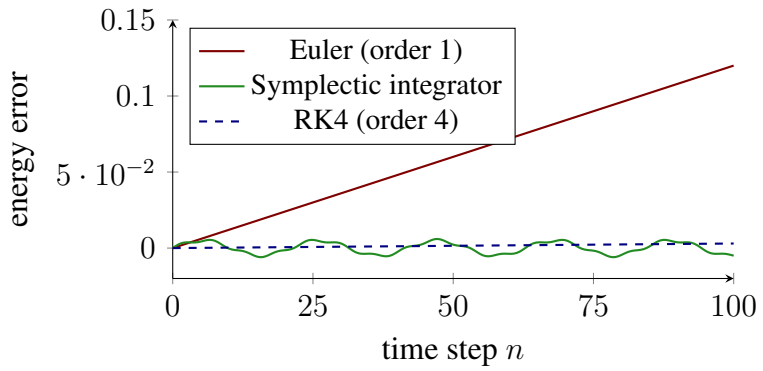


Figure 16.1: Energy error over time for different integrators applied to a Hamiltonian system. Euler accumulates linear drift; symplectic integrators exhibit bounded oscillatory error; RK4 drifts slowly.

16.3 Conservation-Preserving Methods

Symplectic integrators preserve the symplectic structure of Hamiltonian systems. Projection methods enforce constraints:

$$x_{n+1} \leftarrow \Pi_M(x_{n+1}).$$

These methods explicitly incorporate geometric structure, making invariant preservation a design feature rather than a lucky accident.

16.4 Modelling as Layered Representation

The structure of any simulation has three layers:

1. **Symbolic layer**: $\dot{x} = F(x)$ — the geometry.
2. **Numerical layer**: $x_{n+1} = \tilde{\Phi}_h(x_n)$ — the approximation.
3. **Implementation layer**: executable code — the encoding.

Structural integrity requires each layer to remain aligned with the one above it. When the implementation diverges from the numerical scheme, artefacts appear. When the numerical scheme fails to approximate symbolic structure faithfully, qualitative errors emerge.

Chapter 17

Retinal Cell as Dynamical System

17.1 Membrane Potential as a State Variable

A retinal photoreceptor or ganglion cell is an excitable dynamical system. Current balance gives:

$$C\frac{dV}{dt} = -\sum_i I_i(V, \mathbf{w}) + I_{\text{stim}}(t),$$

where C is membrane capacitance, $I_i = g_i(\mathbf{w})(V - E_i)$ are ionic currents with conductance g_i and reversal potential E_i, and $\mathbf{w}$ denotes gating variables.

17.2 Gating Variables as Coupled Dynamics

Each gating variable w satisfies

$$\frac{dw}{dt} = \frac{w_\infty(V) - w}{\tau_w(V)},$$

where $w_\infty(V)$ is the voltage-dependent steady state and $\tau_w(V)$ the voltage-dependent time constant. The full state vector is $x = (V, w_1, w_2, \dots)^\top$, and the system is $\dot{x} = F(x, t)$.

17.3 Phototransduction as Nested Dynamics

In photoreceptors, light L modifies cGMP concentration $c(t)$:

$$\frac{dc}{dt} = -k_{\text{hyd}}(L)\, c + k_{\text{syn}}(c),$$

and channel conductance depends on c:

$$g_{\text{light}}(c) = g_{\max} \frac{c^n}{c^n + K^n}.$$

The compositional chain is:

$$L \xrightarrow{\text{chem}} c(t) \xrightarrow{g(c)} I(V, c) \xrightarrow{\text{current balance}} V(t).$$

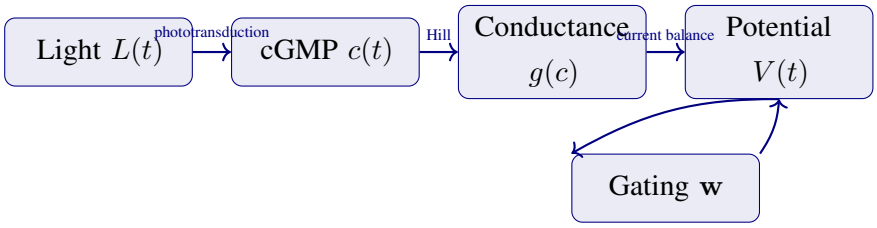


Figure 17.1: Functional decomposition of the retinal photoreceptor. Each arrow is a morphism between state spaces; derivatives propagate via the chain rule.

17.4 Functional Decomposition

The total vector field decomposes as:

$$F_{\text{retina}} = F_{\text{membrane}} + F_{\text{gating}} + F_{\text{photo}}.$$

Each component is a pure function $X \to TX$. Composition is algebraic addition. Subsystems remain modular because they are functions, not procedural updates.

17.5 Linearisation and Stability

At equilibrium x_0 under constant illumination,

$$J(x_0) = DF(x_0) = \begin{pmatrix} \partial F_V/\partial V & \partial F_V/\partial \mathbf{w} \\ \partial F_{\mathbf{w}}/\partial V & \partial F_{\mathbf{w}}/\partial \mathbf{w} \end{pmatrix}.$$

Eigenvalues of $J(x_0)$ determine excitability, adaptation speed, and resonance. All these are spectral properties of a single matrix.

Chapter 18

Skin Cell and Boundary Regulation

18.1 Epidermal Dynamics as Reaction–Diffusion System

Unlike the retinal neuron, the skin cell operates within a layered tissue architecture governed by chemical gradients and boundary constraints. Let $c(x, t)$ denote signalling molecule concentration in region Ω:

$$\frac{\partial c}{\partial t} = D\,\Delta c + R(c).$$

The Laplacian Δc measures spatial curvature of the concentration field. Diffusion smooths gradients; reaction terms may amplify or suppress concentration locally.

18.2 Boundary Conditions as Structural Constraints

The basement membrane at $\partial\Omega$ imposes boundary conditions:

- Dirichlet: $c\big|_{\partial\Omega} = c_0$ (fixed concentration).
- Neumann: $\partial c/\partial n\big|_{\partial\Omega} = 0$ (no flux).

These restrict the state to a subspace $\mathcal{F}_{\text{admissible}} \subset \mathcal{F}$. Constraint again appears as geometric restriction—now on infinite-dimensional function space.

18.3 Turing Instability and Pattern Formation

Theorem 18.1 Turing instability

Consider a two-component reaction–diffusion system near a homogeneous steady state (c_0, m_0) stable in the absence of diffusion. If the diffusion coefficients $D_c \neq D_m$ satisfy a certain parameter inequality, spatial modes of the Laplacian with wavenumber k^* can become unstable, producing spontaneous patterning.

Eigenvalue analysis of the spatially extended Jacobian (incorporating $-Dk^2$ terms from diffusion) determines which spatial frequencies grow. Pattern formation emerges when reaction kinetics introduce positive feedback exceeding diffusion smoothing.

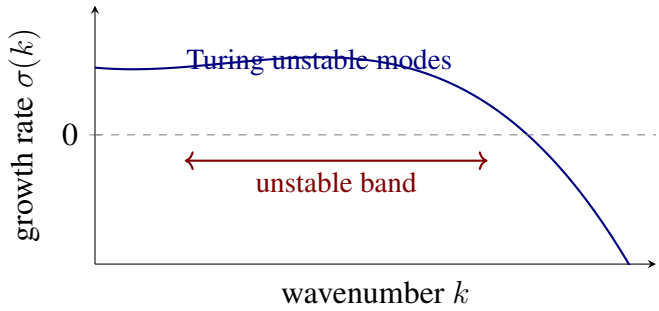


Figure 18.1: Dispersion relation for a Turing system. Modes in the unstable band (Maroon) have positive growth rate $\sigma(k) > 0$ and produce spatial patterns.

18.4 Fibrosis as Bifurcation in Repair Dynamics

A minimal wound-repair model:

$$\frac{dm}{dt} = k_1 c + k_3 m^2 - k_2 m.$$

When repair gain k_3 exceeds a threshold, the steady state of low m loses stability and excessive matrix accumulation occurs. Pathology corresponds to a bifurcation in parameter space—not to a narrative label but to a geometric transition.

Chapter 19

Appliances as Controlled Dynamical Systems

19.1 Thermal System: The Oven Model

Let $T(t)$ be oven temperature:

$$C\frac{dT}{dt} = -k(T - T_{\text{room}}) + P\,u(t),$$

with C thermal capacity, k heat-loss coefficient, P heater power, and $u \in \{0, 1\}$ heater state. The equilibrium temperature under constant input is $T^* = T_{\text{room}} + (P/k)u$.

Under proportional feedback $u = K(T_d - T)$, the closed-loop eigenvalue is $\lambda = -(k + PK)/C$. With $K > 0$, $\lambda < 0$ and the system is exponentially stable.

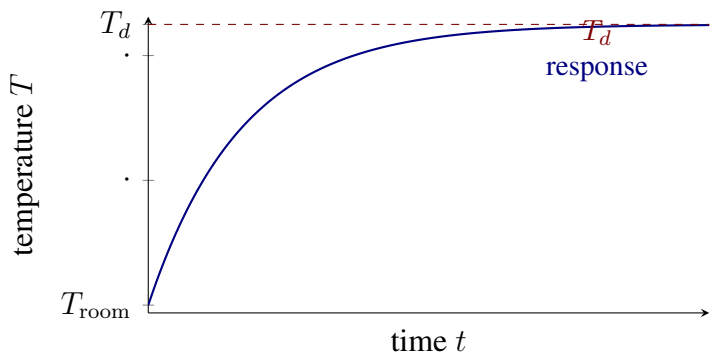


Figure 19.1: Temperature response of an oven under proportional feedback control. The trajectory converges exponentially to the target T_d.

19.2 Washing Machine Drum

Let $x_1 = \theta$ (angle) and $x_2 = \dot{\theta}$ (angular velocity):

$$\frac{dx_1}{dt} = x_2, \qquad \frac{dx_2}{dt} = \frac{1}{I}(\tau_{\text{motor}}(u) - bx_2 - \tau_{\text{load}}).$$

Spin cycles, agitation patterns, and acceleration ramps are trajectories in this two-dimensional state space. Control selects the trajectory.

19.3 Compositional Architecture

Each subsystem is a pure function:

$$F_{\text{total}} = F_{\text{thermal}} + F_{\text{mechanical}} + F_{\text{control}}.$$

Subsystems remain modular because they map state to tangent direction. The system is declarative rather than procedural. Safety constraints define invariant sets; control ensures trajectories remain inside them.

Chapter 20

A Functional Simulation Architecture

20.1 State as a Structured Object

Define state as a single coherent element of configuration space:

$$x = (V, w_1, w_2, T, c_1, \dots)^\top \in X.$$

State is not scattered across global variables.

20.2 Pure Vector Field Definition

Define dynamics as a pure function $F : X \to TX$:

F(x) - pure vector field

```
function F(x, t, params):
    return {
        dV  = membrane(x, params),
        dw1 = gating1(x, params),
        dw2 = gating2(x, params),
        dT  = thermal(x, params),
        dc1 = chemistry(x, params)
    }
```

Each component depends only on the input state. No mutation occurs inside F. The

function is referentially transparent.

20.3 Integrator as Separate Layer

step_rk4 - 4th-order Runge-Kutta

```
function step_rk4(F, x, t, h, params):
    k1 = F(x,           t,       params)
    k2 = F(x + h/2*k1,  t + h/2, params)
    k3 = F(x + h/2*k2,  t + h/2, params)
    k4 = F(x + h*k3,    t + h,   params)
    return x + h/6 * (k1 + 2*k2 + 2*k3 + k4)

# Simulation loop
x = x0;  t = 0
for k = 1..N:
    x = step_rk4(F, x, t, h, params)
    t = t + h
```

20.4 Constraint Projection as Operator

project - enforce admissible manifold

```
function project(x):
    # Enforce mass conservation
    correction = (sum(x.mass) - M_total) / N
    x.mass = x.mass - correction
    # Enforce non-negativity
    x.concentrations = max(x.concentrations, 0)
    return x

# Modified simulation loop
for k = 1..N:
    x = step_rk4(F, x, t, h, params)
    x = project(x)          # enforce constraints
    t = t + h
```

20.5 Alignment with Calculus

Mathematics	**Code**	**Role**
State manifold X	Structured type / record	Configuration space
Vector field $F : X \to TX$	Pure function	Instantaneous rate
Flow Φ_t	`step_rk4` loop	Time evolution
Projection Π_M	`project` function	Constraint enforcement
Jacobian J_F	Automatic differentiation	Sensitivity analysis
Eigenvalues of J_F	Stability check	Qualitative behaviour

Chapter 21

Worked Example I: Minimal Retinal Cell

21.1 The FitzHugh–Nagumo Model

A minimal excitable model for a retinal ganglion cell uses the FitzHugh–Nagumo system. State: $x = (V, n)^\top$ where V is membrane potential and n is a recovery variable. Light input enters as external current $I(t)$:

$$\frac{dV}{dt} = \underbrace{V - \frac{V^3}{3}}_{\text{fast nonlinearity}} - n + I(t), \qquad \frac{dn}{dt} = \varepsilon(V + a - bn).$$

21.2 Jacobian and Local Sensitivity

The Jacobian is explicitly:

$$J(x, t) = \begin{pmatrix} 1 - V^2 & -1 \\ \varepsilon & -\varepsilon b \end{pmatrix}.$$

This matrix is not ornamental. Its eigenvalues determine whether the cell fires repetitively or returns to rest after a perturbation. Small ε (slow recovery) produces action-potential-like spikes; large ε produces damped oscillations.

21.3 Numerical Simulation

FitzHugh–Nagumo implementation

```
# Parameters: a=0.7, b=0.8, eps=0.08
# Input: I(t) = 0.5 (constant stimulus)

function F_retina(x, t, p):
    V = x.V;  n = x.n
    dV = V - (V^3)/3 - n + I(t)
    dn = p.eps * (V + p.a - p.b * n)
    return {dV=dV, dn=dn}
```

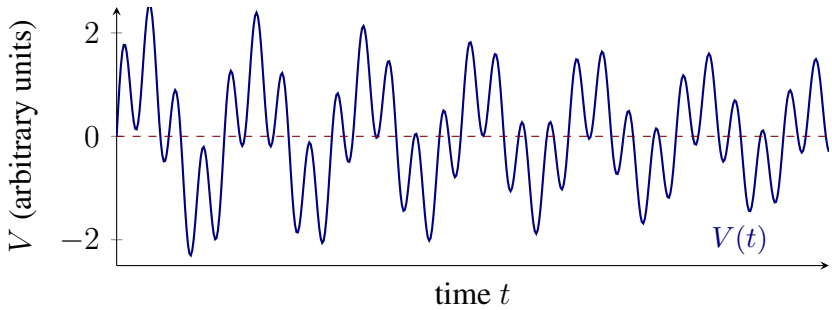


Figure 21.1: Schematic voltage trace of the FitzHugh–Nagumo model under constant stimulus. The oscillatory behaviour arises from the interplay of fast voltage nonlinearity and slow recovery variable.

21.4 Control Interpretation

If $I(t)$ is a control input rather than a fixed stimulus, the retinal cell is a control system:

$$\dot{x} = F_0(x) + G\,u(t), \qquad G = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad u(t) = I(t).$$

Photonic input actuates only the voltage direction. The slow variable n is not directly actuated—it responds only through coupling.

Chapter 22

Worked Example II: Skin Cell and Boundary Repair

22.1 Minimal Repair Model

State: $x = (B, W)^\top$ where $B \in [0, 1]$ is basement membrane integrity and $W \geq 0$ is wound burden.

$$\frac{dW}{dt} = u(t) - \alpha\, B\, W - \gamma\, W,$$
$$\frac{dB}{dt} = r(t)\, \beta\, W - \delta\, B + \eta\, B(1 - B).$$

Wound burden increases with injury $u(t)$, decreases when membrane integrity suppresses disruption, and decays through baseline clearance. Basement membrane integrity increases when repair resources $r(t)$ respond to wound burden, decays through turnover, and saturates through logistic self-regulation.

22.2 Phase Plane and Equilibrium Analysis

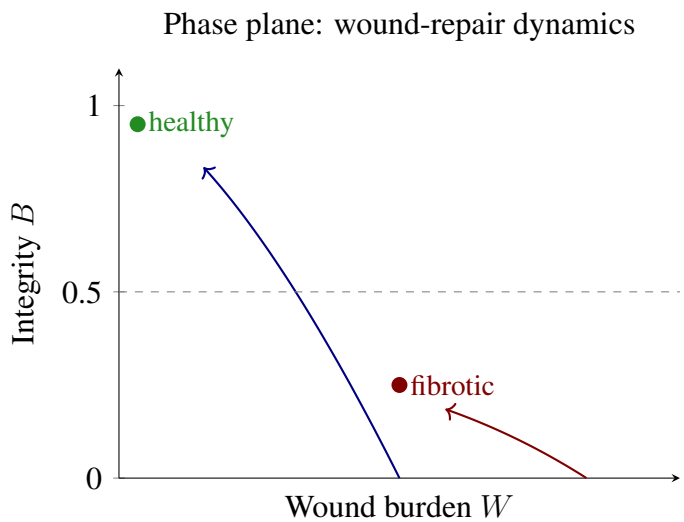


Figure 22.1: Phase plane for the wound-repair model. Trajectories either converge to the healthy equilibrium (high integrity, low wound burden) or the fibrotic state (low integrity, sustained wound burden), depending on initial conditions and parameter regime.

22.3 Projection Enforcement

Since $B \in [0, 1]$ is a physical constraint:

project_skin - admissible manifold

```
function project_skin(x):
    B = clamp(x.B, 0.0, 1.0)
    W = max(x.W, 0.0)
    return {B=B, W=W}
```

Simulation:

```
for k = 1..N:
    x = step_rk4(F_skin, x, t, h, params)
    x = project_skin(x)    # enforce [0,1] x R+
    t = t + h
```

The constraint is part of the *architecture*, not a numerical afterthought.

Chapter 23

Manifold-Aligned Inference and the Discipline of Not Predicting Noise

23.1 From Dynamical Systems to Inference

An inference system also evolves state. The state is now a belief, parameter estimate, or internal model; data acts as input; update rules define motion in hypothesis space:

$$\frac{d\theta}{dt} = F(\theta, \text{data}).$$

The same calculus architecture applies.

23.2 Signal and Noise as Geometric Decomposition

Observed data $y = f(x) + \varepsilon$ can be decomposed with respect to the data manifold M:

$$y = \Pi_M(y) + \Pi_N(y),$$

where Π_M projects onto M (structure) and Π_N onto its normal complement (noise).

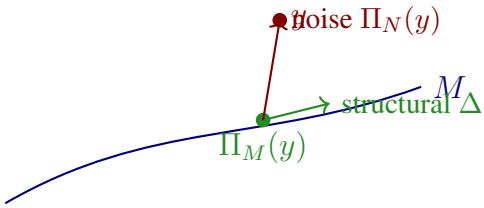


Figure 23.1: Geometric decomposition of a data point y into its tangential component (structure, green) and normal component (noise, red) relative to the data manifold M.

23.3 Curvature Governs Generalisation

The Hessian $H(\theta) = \nabla^2 L(\theta)$ of the loss plays the role of curvature in parameter space. Directions of large positive curvature correspond to stiff, sensitive directions. If those directions correspond to noise modes, aggressive descent chases unstable non-generalisable structure.

Key Insight

The injunction "never predict noise" is a curvature discipline. It suppresses parameter motion aligned with transient, non-structural curvature in the loss landscape — identical in principle to projecting vector fields onto constraint manifolds.

23.4 Noise Characterisation

Noise directions are characterised by:

- Lack of persistence across scales.
- Absence of stable eigenstructure.
- Sensitivity to minor perturbation.
- Failure to generalise under refinement.

These criteria mirror the definition of limit: *structure is what survives refinement; noise vanishes under projection and smoothing*.

23.5 Regularisation as Geometric Constraint

Regularisation modifies the loss:

$$L_\lambda(\theta) = L(\theta) + \lambda R(\theta).$$

For quadratic $R(\theta) = \|\theta\|^2$, the effective Hessian becomes $H_\lambda = H + \lambda I$. Regularisation increases curvature uniformly, shrinking flat directions and improving conditioning. It is feedback control in parameter space, biasing trajectories toward low-curvature, stable regions.

Chapter 24

Entropy, Information, and Curvature in Model Space

24.1 The Fisher Information Metric

For a parametric family $\{p(x;\theta)\}$, the Fisher information matrix

$$\mathcal{I}(\theta) = \mathbb{E}\big[\nabla_\theta \log p(x;\theta) \nabla_\theta \log p(x;\theta)^\top\big]$$

acts as a Riemannian metric on parameter space [25]. Model space is not flat; it carries curvature induced by likelihood geometry.

24.2 Free Energy as Variational Functional

Variational inference minimises the free energy:

$$\mathcal{F}(q) = \mathbb{E}_q[\log q(\theta)] - \mathbb{E}_q[\log p(D,\theta)].$$

This balances entropy of belief against data fit — identical in structure to the energy functionals studied in variational calculus. Inference becomes constrained minimisation on a functional manifold.

24.3 Stability Requires Curvature–Entropy Alignment

- High entropy without curvature $\Rightarrow$ diffuse ignorance.
- High curvature without entropy $\Rightarrow$ brittle overconfidence.
- Stable inference requires alignment: curvature along persistent directions, entropy retained along indeterminate directions.

This is identical to stability in physical systems: stiffness without damping leads to oscillation; damping without structure leads to collapse.

Chapter 25

The Unified Geometry of Physiology, Engineering, and Inference

25.1 One Architecture, Three Domains

The mathematical architecture governing physiological regulation, engineered systems, and inferential processes is formally identical. The differences lie in substrate and interpretation, not in underlying mathematical form.

Concept	**Physiology**	**Engineering**	**Inference**
State manifold	Membrane + channel	Temperature + pressure	Parameter space
Vector field	Ionic kinetics	Mechanical laws	Gradient flow
Constraints	Charge neutrality	Safety bounds	Normalisation
Projection	Conservation	Constraint projection	Regularisation
Curvature	Hessian of energy	Jacobian of F	Fisher information
Control	Neural modulation	Feedback controller	Learning rate
Stability	Lyapunov function	Eigenvalue placement	Loss landscape

25.2 A Single Geometric Doctrine

Each system is defined by an explicit state manifold. Evolution is determined by a vector field. Constraints restrict motion through projection. Stability is governed by curvature. Control modifies trajectories through admissible inputs. Entropy measures dispersion

relative to invariant structure. Coherence persists where curvature and constraint organise flow into stable manifolds.

Key Insight

These principles apply without alteration whether the substrate is biological tissue, mechanical apparatus, or parameter space of a statistical model. The geometry is the same; only the interpretation differs.

Chapter 26

Models, Manifolds, and Adequacy

26.1 What a Model Is

A mathematical model is a structured mapping $f : X \to Y$ between state spaces. The central question is not whether a model is true, but whether it is *adequate* for the structure under consideration.

26.2 Dimensionality and the Curse

If the true system dynamics lie on a k-dimensional submanifold $M \subset \mathbb{R}^n$ with $k \ll n$, the learning problem has effective dimension k. When M does not exist—when variance is evenly distributed across many directions—no low-dimensional approximation is faithful. The curse of dimensionality is the geometric statement that volume concentrates near the boundary of high-dimensional balls, making uniform sampling exponentially sparse.

26.3 Limits of Fixed-Phase-Space Modelling

Classical differential modelling presupposes a fixed phase space. If the underlying system modifies its own variables, the effective manifold X_t evolves. When no smooth diffeomorphism maps X_t into a fixed reference manifold, a time-invariant model cannot capture the system globally.

Key Insight

The limits of modelling are not matters of optimism or pessimism. They are matters of geometry. When phase spaces evolve, dimensions shift, or generative mechanisms alter, no fixed parametric family can guarantee structural fidelity.

Chapter 27

Ergodicity and the Limits of Sampling

27.1 The Ergodic Theorem

Theorem 27.1 Birkhoff's Ergodic Theorem

Let (X, μ, ϕ_t) be a measure-preserving dynamical system and $f \in L^1(\mu)$. Then for μ-almost every x,

$$\lim_{T\to\infty} \frac{1}{T} \int_0^T f(\phi_t(x))\, dt = \int_X f\, d\mu.$$

Ergodicity asserts that time exploration of a single trajectory suffices to recover global statistical structure. This property is not automatic; it is a structural constraint on the dynamics.

27.2 Non-Ergodicity and Phase-Space Fragmentation

If phase space decomposes into invariant components $X = \bigcup_\alpha X_\alpha$ with trajectories confined within each X_α, then time averages depend on initial condition. No amount of additional time reveals structure in other components. Non-ergodicity imposes a fundamental geometric limit on inference.

27.3 Stationarity and Distributional Drift

Ergodicity presupposes stationarity: the law governing $x(t)$ is invariant under time translation. If system parameters drift or external forcing varies, time averages no longer converge to a fixed ensemble average. Learning from past data presumes persistence of generative structure.

Chapter 28

The Absolute Limits of Approximation

28.1 Approximation as Projection onto a Hypothesis Manifold

Every parametric model defines a hypothesis manifold $\mathcal{H}$ within function space. Approximation projects data onto $\mathcal{H}$ by minimising loss. If the true generative mapping lies near $\mathcal{H}$, projection yields small residual. If not, approximation error persists regardless of optimisation precision.

28.2 Five Geometric Limits

1. **Fixed-dimensionality assumption**. When dimensionality evolves, no fixed Θ remains adequate.
2. **Stationarity assumption**. When P_t drifts, minimising empirical risk converges to outdated structure.
3. **Bounded-perturbation assumption**. In chaotic regimes, small perturbations produce large deviations; approximation requires exponentially increasing precision.
4. **Ergodicity assumption**. Sampling fails to represent invariant measure if phase space is fragmented.
5. **Curvature mismatch**. Local agreement does not imply global fidelity; second-order geometry must align.

Key Insight

Increasing computational power accelerates exploration of Θ but does not alter the geometry of $\mathcal{H}$ or the true generative manifold. Approximation limits arise from geometry, not computation.

28.3 Where Curvature Is Stable, Refinement Succeeds

When the five assumptions hold, calculus guarantees convergence under refinement. When they fail, no increase in computation ensures structural fidelity.

Calculus defines both the power and the boundary of approximation. Where curvature is stable, refinement succeeds. Where geometry shifts, projection fails. The boundary is written in the structure of phase space itself.

Bibliography

[1] W. Rudin. *Principles of Mathematical Analysis*. McGraw–Hill, 3rd edition, 1976.

[2] W. Rudin. *Real and Complex Analysis*. McGraw–Hill, 3rd edition, 1987.

[3] G. B. Folland. *Real Analysis: Modern Techniques and Their Applications*. Wiley, 2nd edition, 1999.

[4] L. C. Evans. *Partial Differential Equations*. American Mathematical Society, 2nd edition, 2010.

[5] J. M. Lee. *Introduction to Smooth Manifolds*. Springer, 2nd edition, 2012.

[6] J. M. Lee. *Riemannian Manifolds: An Introduction to Curvature*. Springer, 1997.

[7] M. P. do Carmo. *Riemannian Geometry*. Birkhäuser, 1992.

[8] R. Abraham, J. E. Marsden, and T. Ratiu. *Manifolds, Tensor Analysis, and Applications*. Springer, 2nd edition, 1988.

[9] V. Guillemin and A. Pollack. *Differential Topology*. Prentice–Hall, 1974.

[10] M. W. Hirsch, S. Smale, and R. L. Devaney. *Differential Equations, Dynamical Systems, and an Introduction to Chaos*. Academic Press, 3rd edition, 2012.

[11] A. Katok and B. Hasselblatt. *Introduction to the Modern Theory of Dynamical Systems*. Cambridge University Press, 1995.

[12] S. H. Strogatz. *Nonlinear Dynamics and Chaos*. Westview Press, 2nd edition, 2015.

[13] J. Guckenheimer and P. Holmes. *Nonlinear Oscillations, Dynamical Systems, and Bifurcations of Vector Fields*. Springer, 1983.

[14] S. Wiggins. *Introduction to Applied Nonlinear Dynamical Systems and Chaos*. Springer, 2nd edition, 2003.

[15] P. Walters. *An Introduction to Ergodic Theory*. Springer, 1982.

[16] I. P. Cornfeld, S. V. Fomin, and Y. G. Sinai. *Ergodic Theory*. Springer, 1982.

[17] P. Billingsley. *Probability and Measure*. Wiley, 3rd edition, 1995.

[18] R. Durrett. *Probability: Theory and Examples*. Cambridge University Press, 4th edition, 2010.

[19] E. D. Sontag. *Mathematical Control Theory*. Springer, 2nd edition, 1998.

[20] H. K. Khalil. *Nonlinear Systems*. Prentice Hall, 3rd edition, 2002.

[21] A. Isidori. *Nonlinear Control Systems*. Springer, 3rd edition, 1995.

[22] J.-L. Lions. *Optimal Control of Systems Governed by Partial Differential Equations*. Springer, 1971.

[23] R. W. Brockett. Asymptotic stability and feedback stabilization. In R. W. Brockett, R. S. Millman, and H. J. Sussmann, editors, *Differential Geometric Control Theory*, pages 181–191. Birkhäuser, 1983.

[24] T. M. Cover and J. A. Thomas. *Elements of Information Theory*. Wiley, 2nd edition, 2006.

[25] S. Amari and H. Nagaoka. *Methods of Information Geometry*. American Mathematical Society, 2000.

[26] V. N. Vapnik. *Statistical Learning Theory*. Wiley, 1998.

[27] C. M. Bishop. *Pattern Recognition and Machine Learning*. Springer, 2006.

[28] T. Hastie, R. Tibshirani, and J. Friedman. *The Elements of Statistical Learning*. Springer, 2nd edition, 2009.

[29] S. Boyd and L. Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004.

[30] H. B. Callen. *Thermodynamics and an Introduction to Thermostatistics*. Wiley, 2nd edition, 1985.

[31] J. E. Marsden and T. S. Ratiu. *Introduction to Mechanics and Symmetry*. Springer, 2nd edition, 1999.

[32] M. Wilson. *Physics Avoidance and Other Essays in Conceptual Strategy*. Oxford University Press, 2017.

[33] N. Cartwright. *How the Laws of Physics Lie*. Oxford University Press, 1983.